

Khawaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Mid-term Examination: Spring Semester 2020

Program: B.Sc. in CSE (Batch – 10th) 1st Year 1st Semester

Course Title: Physics - I

Course Code: PHY 1101

Time: 15 Minutes

Total Marks: 15 X 1 = 15

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

- The potential differences between two points in an equipotential surface is
a) 0 b) 1 c) 2 d) 3 e) 4
- What is the unit of the relative permittivity?
a) *Volt* b) *Joule* c) *C* d) $C^2 N^{-1} m^{-2}$ e) *None*
- The net flux of a cylindrical Gaussian surface placed in an external field is
a) negative b) positive c) infinity d) maximum e) zero
- Conductance is to the cross-sectional area of the conductor.
a) equal b) independent c) reciprocal d) proportional e) inversely proportional
- What is the value of the minimum amount of charge in universe?
a) $1.6 \times 10^{-16} C$ b) $1.6 \times 10^{-19} C$ c) $9.1 \times 10^{-31} C$ d) $1.9 \times 10^{-16} C$ e) $3.1 \times 10^{-34} C$
- Which one is the form of Ohm's law?
a) $E = \rho J$ b) $J = \rho E$ c) $F = \rho J$ d) $J = \rho F$ e) $E = JA$
- The unit of the electric flux is
a) $N m^2 C^{-1}$ b) $C^2 N^{-1} m^{-2}$ c) $N m^{-2}$ d) $C m^{-2}$ e) $m^2 C^{-1} N^{-2}$
- If the surface does not enclose the charge, then the $\oint_S \vec{E} \cdot \vec{da} = ?$
a) $\frac{\epsilon_0}{q}$ b) $\frac{q}{\epsilon_0}$ c) $\epsilon_0 q$ d) ϵ_0 e) 0
- What is the permittivity of the medium of dielectric constant 1.5?
a) $1.5 \times 10^{-12} F m^{-1}$ b) $8.85 \times 10^{-12} F m^{-1}$ c) $13.3 \times 10^{-12} F m^{-1}$ d) $15.5 \times 10^{-12} F m^{-1}$ e) $20.1 \times 10^{-12} F m^{-1}$
- What is the value of $\frac{1}{4\pi\epsilon_0}$?
a) $1.6 \times 10^{-16} N m^{-1}$ b) $9 \times 10^9 N m^2 C^{-2}$ c) $8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$ d) $6.6 \times 10^{-16} N m^{-1}$ e) $1.6 \times 10^{-19} N m^{-1}$
- The tangent of any point in the electric field lines indicates the
a) current. b) voltage. c) emf. d) resistance. e) electric field intensity.
- What is the equivalent of Ampere unit?
a) AC^{-1} b) NC^{-1} c) CN^{-1} d) CA^{-1} e) CF^{-1}
- $\int \vec{J} \cdot \vec{da} = ?$
a) V b) E c) I d) R e) C
- Which one is the right equation?
a) $V = -\nabla E$ b) $\vec{E} = -\nabla V$ c) $\vec{F} = -\nabla V$ d) $F = -\nabla V$ e) $E = -\nabla R$
- What is the unit of electric potential energy?
a) Joule b) Volt c) Farad d) Ohm e) Volt-meter

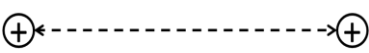
Part 2: SAQ

Time: 45 Minutes

Total Marks: 15

[Answer any 03 (three) questions from the followings]

[The figures in the right margin indicate full marks for the respective question]

- What is electrostatics? [1]
 - How can you stay safe during a lightning storm? [4]
- Establish a relationship between current density & charge density. [2]
 - The parallel plates of an air-filled capacitor are everywhere 1.0 mm apart. What must be the plate area if the capacitance is to be 1.0 farad? [3]
- Is it possible to charge a capacitor with any potential? – justify your answer. [2]
 - $q_1 = 1.0 \times 10^{-6} C$ $q_2 = 2.0 \times 10^{-6} C$

 $r = 10 cm$
At what point the line joining the above two charges is the electric field strength zero? [3]
- Define dielectric constant. [2]
 - What is the effect of dielectric materials in the capacitor? – Explain. [3]